RingClock & BitClock

A 60-LED Animated Ring · A Binary BCD Strip

User's Guide

Oct 2026

This combined guide covers the setup, configuration, and day-to-day operation of two Angry Electrons clocks: the RingClock — a 60-LED addressable ring with ten display modes plus Cycle — and the BitClock — a binary-coded-decimal display rendered on a configurable APA102 LED strip. Both clocks share the same WiFi setup, NTP time sync, web settings, scheduled rules, and over-the-air update mechanism, so the platform chapters apply equally to both. Display-specific chapters are clearly labeled.

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Designed by Mitchell Feig — Angry Electrons
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Quick Start

The RingClock and the BitClock share the same Angry Electrons software platform, but they look completely different on the outside. The steps in this Quick Start work for both — sections that apply to one model only are called out clearly. A full explanation of every option is in the numbered sections that follow.

1. Power on

Connect a 5 V USB-C power supply. On a brand-new clock, AP setup mode starts automatically — see step 3.

BitClock note: an LED strip longer than 100 LEDs draws more current than the USB-C input can safely supply on its own. For strips above 100 LEDs, connect an external 5 V supply directly to the strip's power injection points and use the USB-C only for the controller; otherwise the far end of the strip will dim, flicker, or shift color under load.

2. Wait for the visual cue (RingClock only)

On a freshly programmed RingClock, you may need to confirm the install orientation before the time reads correctly. If the boot version digits read backwards, mirrored, or scroll the wrong way, see Section 5: Power-On Functions for the orientation-cycle procedure. This is a one-time setup step at install time.

3. Connect to the clock's WiFi

From a phone or computer, open the WiFi settings and look for the network named after the clock — RingClock or BitClock by default (or whatever Access Point / mDNS Name is configured). Connect to it. No password is required for the AP.

4. Configure in a browser

Open a web browser and go to 192.168.4.1. The setup page will load. At a minimum, do the following:

- Select the home WiFi network from the SSID dropdown and enter its password. On a clock with no saved network the list starts on "-- select your network --"; select the home network before pressing Start.
- Confirm the UTC offset for your local region (US Eastern is 5 hours, Minus, by default).

On the BitClock, also confirm LED Length matches the strip (default 33; maximum 500). Tap Save, then Start. The clock saves the settings and reboots into live mode.

5. Wait for time to appear

On a successful boot, the clock connects to WiFi and fetches time from an NTP server. Each clock signals progress in its own way:

- RingClock — version digits flash briefly at 12 o'clock, then the time renders in the configured display mode.
- BitClock — a short colored sweep travels along the strip: green while connecting to WiFi, blue while syncing NTP, then the time renders as BCD groups.

If WiFi association and the initial NTP sync have not both completed within about two minutes, the clock reboots and retries.

After three failed boots in a row (for example a wrong WiFi password or a renamed network), the next boot starts the setup access point instead, so the WiFi settings can be corrected at 192.168.4.1. Because each attempt allows about two minutes, this takes several minutes. This forced AP mode lasts for that one boot only: correct the settings and press Start to save them and reboot into normal operation. Pressing Save without Start stores the settings but does not keep the clock in AP mode; the next power-cycle tries the saved network again. If the network was only temporarily down (router rebooting, internet outage), power-cycling the clock returns it to normal operation.

6. Daily use

To do this	Do this
See the clock's IP address (RingClock)	Double-click SW1 — the IP renders on the ring; click SW1 again to dismiss
Adjust most any setting live	Browse to the clock's IP address (or http://<ap-name>.local/) from any device on the same WiFi
Cycle BitClock brightness	Click SW2 — steps 1 → 2 → … → 9 → 1, persisted immediately
Re-enter AP configuration mode	RingClock: release SW2 after 5–10 seconds; BitClock: hold 5 seconds
Install new software over WiFi	Hold SW1 for 10 seconds while the time is displayed
Factory reset	Hold SW1 + SW2 while applying power; clears settings and rules, then enters AP setup

7. Recommended next steps

- Open the live settings page and explore the Display Mode and Periodic Animation controls (RingClock) or time and date colors (BitClock) without committing — every change applies in real time, only Save persists it.
- Set up Scheduled Rules to dim or change settings during sleeping hours.
- Confirm the OTA URL is set so future software releases can install with a single button hold.

1. Overview

This guide covers two clocks built on the same Angry Electrons software platform: the RingClock — a 60-LED addressable ring with ten display modes plus Cycle — and the BitClock — a binary-coded-decimal display rendered on an APA102 LED strip of configurable length. They look nothing alike, but underneath they share the same WiFi setup, NTP time sync, web-based live settings, scheduled rules, and over-the-air update mechanism.

What is a RingClock?

The RingClock is a 60-LED ring that displays time as analog hour, minute, and second hands — each independently colorable, each drawn in the selected display mode. Ten display modes range from minimalist single-LED dots through comet trails, orbits, sonar-style radar sweeps, gravity drops, and spark chases. Optional always-on layers (5-minute marks, quarter marks, a sweeping pendant, and a slow-swinging pendulum) fill in the rest of the dial.

What is a BitClock?

The BitClock displays time as BCD (binary-coded decimal). Six four-bit groups represent HH:MM:SS, drawn as lit and unlit LED blocks along a strip. Four sliders on the setup LED Layout page tune block width, colon size, and inter-block spacing so the layout can be matched to the exact length of a strip from 33 LEDs up to 500. Four independently-set colors render the lit bits, the unlit bits, the colons, and the background.

Shared Platform Features

- WiFi setup with a 2.4 GHz network (5 GHz networks are not supported by the ESP32).
- UTC offset including 30- and 45-minute time zones.
- Automatic Daylight Saving Time start and end with configurable rules.
- NTP time synchronization with a configurable primary and alternate server.
- Web-based live settings page accessible at the clock's IP or via mDNS (<ap-name>.local).
- Time-range scheduled rules with their own brightness and (per clock) display settings.
- Over-the-air software updates triggered by a 10-second hold of SW1.
- Optional Live Form Password protecting the live settings page.
- Factory reset by holding both buttons during power-on.

2. Models

Both clocks ship as either DIY kits or fully-assembled units. The variant choice does not affect the software — the same firmware runs on every unit. The RingClock is available as either a small or large version.

Model	Display	Default LED Count	Power
RingClock	60-LED ring (APA102 or WS2812B)	60 (fixed)	5 V USB-C
BitClock	APA102 LED strip	33 (configurable, max 500)	5 V USB-C up to 100 LEDs; external 5 V injection required for longer strips

The RingClock's ring count is fixed at 60 and runs entirely from the USB-C input. The BitClock's strip length is configured during AP setup; choose a value that matches the physical strip. Strips of 100 LEDs or fewer can be powered from the USB-C supply alone. Above 100 LEDs, the cumulative current draw exceeds what USB-C can deliver — feed an external 5 V supply directly into the strip's injection points and let USB-C power the controller only.

3. Display

RingClock — Hands and Layers

The RingClock renders three independent "hands" on the 60-LED ring: hour, minute, and second. Each hand has its own color. The ten display modes plus Cycle (Section 7) determine how the hands draw themselves: as single dots, comet trails, orbits, radar sweeps, wedges, and so on.

Seven effect layers can be enabled independently and overlaid on whichever display mode is selected.

- Pendulum — a bob that swings slowly between the 20- and 40-minute positions (centered at 6 o'clock) using simple harmonic motion — slow at the extremes, fast through the bottom of the arc, like an actual pendulum. Three-second full back-and-forth period.
- Pendant — a single dot that swings back and forth across the entire ring, leaving a fading trail.
- 5-Minute Marks — twelve LEDs at every fifth position around the ring, like the major markings on an analog clock face.
- Quarter Marks — four LEDs at the 0, 15, 30, and 45 positions (12, 3, 6, 9 o'clock). Layered on top of the 5-minute marks.
- Sparkle — random LEDs flash in the configured Sparkle color and decay over ~250 ms each, about one new sparkle per second on average. Adds shimmer without obscuring the hands.
- Background — a constant dim wash across all 60 LEDs in the configured Background color, useful as a base over which the hands and other effects layer.
- Drift — an 8-LED comet trail drifting counter-clockwise around the ring at roughly 1 LED per second in the configured Drift color.

Setting any of these to all-zero (black) disables that layer.

A separate Color Rotation toggle slowly cycles the hue of every configured color — hands, pendulum, pendant, marks, sparkle, background, and drift — around the color wheel in lockstep; a Rotation Period slider sets how many minutes one full hue cycle takes (1–360, default 60). Disabling rotation snaps every color back to its picked value with no precision loss. Every effect also carries its own brightness slider (0–9) directly beneath its color, so any one effect can be dimmed without dimming the hands or the other effects.

BitClock — Reading BCD

Six time digits (HH:MM:SS) appear left-to-right along the strip. Each digit is encoded as four bits, drawn most-significant-bit first. An "on" LED block represents a 1; an "off" block represents a 0. The digit 9 (binary 1001) lights the outer two blocks and leaves the inner two dark; the digit 5 (0101) alternates.

Two colon groups separate hours from minutes and minutes from seconds:

[HH-tens] [HH-ones] : [MM-tens] [MM-ones] : [SS-tens] [SS-ones]

Each bracketed group is four bits wide. Four configurable widths (Section 6) tune the look: how many physical LEDs make up each bit cell, each colon lamp, and each separator. Any LEDs left over after the layout is computed become symmetric margins on the left and right of the strip.

4. Buttons

Both clocks use the same two-button layout: SW1 (left) and SW2 (right). The function of each press depends on the model.

SW1 — During Normal Operation

Action	RingClock	BitClock
Single click	Dismiss the IP-address display	— (no action)
Double-click	Show the clock's IP address on the ring (see Section 12)	— (no action)
Hold 10 seconds	Start an over-the-air software update	Start an over-the-air software update

RingClock also supports forced reinstall: release SW2 after 10+ seconds to install the published release even if its version matches.

SW2 — During Normal Operation

Action	RingClock	BitClock
Single click	— (no action)	Cycle brightness 1 → 2 → … → 9 → 1; persisted immediately
Hold 5–10 seconds and release	Enter AP setup (settings preserved)	Reboot into AP setup mode (settings preserved)

5. Power-On Functions

Special functions run when buttons are held during power-on, before the firmware initializes its normal button handling. Apply power with the buttons held; the clock reads the state at the start of boot.

Factory Reset (Both Clocks)

Hold SW1 and SW2 together while applying power to clear WiFi credentials, display settings and scheduled rules. RingClock enters AP setup during the same boot, without waiting for button release or rebooting again. Its install mode and chip provisioning survive. RingClock defaults are Blend mode, Display Brightness 7, red/green/blue hands and a purple pendulum (#800080) at brightness 5, plus the Hand Spin periodic animation every 1 minute.

Install Mode Cycle RingClock

Hold SW2 alone at power-on to advance the install mode. The choice survives factory reset. Modes 0–3 use physical LED index 29 at 12 o'clock; mode 4 uses index 0. Indexes are zero-based.

Mode	LED type	Direction	Feedback
0 default	APA102	Counterclockwise	1 yellow LED
1	APA102	Clockwise	2 yellow LEDs
2	WS2812B	Counterclockwise	1 cyan LED
3	WS2812B	Clockwise	2 cyan LEDs
4	WS2812B alternate ring	Clockwise, LED 0 at top	3 cyan LEDs

The indicator remains lit until SW2 is released, then startup continues. Repeat power-on with SW2 held to advance 0 → 1 → 2 → 3 → 4 → 0. The default is mode 0.

6. AP Configuration Mode

Both clocks offer AP setup at 192.168.4.1 when unconfigured. On RingClock, release SW2 after 5–10 seconds to return to setup; holding 10 seconds or more requests a forced firmware reinstall.

Entering AP Mode

- Power on a brand-new clock — it starts in AP mode automatically.
- RingClock: hold SW2 for at least 5 but less than 10 seconds, then release, to enter AP setup. A hold of 10 seconds or more requests a forced firmware reinstall.
  BitClock: hold SW2 for 5 seconds to enter setup.
- Apply power with both buttons held to clear settings and scheduled rules and enter AP setup with defaults.
- After three boots in a row that fail to connect to WiFi and sync the time, the next boot starts AP setup for that one boot (see Quick Start, step 5). Press Start after correcting the settings; power-cycling instead retries the saved network.

Connecting

1. Connect a phone or computer to the WiFi network named after the clock — RingClock or BitClock by default. No password is required.

2. Open a web browser and go to 192.168.4.1.

3. Adjust settings (described below) and press Save, then Start. The clock saves all settings and reboots into live mode.

All hourly times use the 24-hour clock.

Setup Pages

RingClock setup has two pages, Setup and Display, plus Start. The Setup page holds every network and time setting, in this order: Network, Static IP Address, UTC Offset, Daylight Saving Time, Standard Time and OTA Update (Firmware URL). The Display page holds the display settings described in Chapter 7.

BitClock setup pages are Setup, Time Settings, LED Layout, Time Display and Date Display, plus Start.

Network Settings (Both Clocks)

Setting	Description
Access Point / mDNS Name	WiFi network name in AP mode; also the mDNS hostname and the DHCP client name shown in the router's device list (lowercased, with non-alphanumerics converted to hyphens). Defaults: "RingClock" or "BitClock". RingClock accepts up to 31 characters.
SSID	Dropdown of scanned 2.4 GHz networks. 5 GHz networks are not supported.
SSID Override / Alternate SSID	Manual text entry that overrides the dropdown when non-empty. Useful for hidden networks or SSIDs containing characters that confuse the dropdown.
Network Password (RingClock) / Password (BitClock)	WiFi password.
NTP Server	Primary time server. RingClock: us.pool.ntp.org. BitClock: time.cloudflare.com.
Alternate NTP Server	Secondary time server. Used as a fallback during boot sync, and rotated to automatically if the primary stops responding for 30 minutes. Default for both clocks: time.google.com. Leave blank to disable rotation.
Live Form Password	Optional password for the live settings web page. Leave blank to disable password protection. On the RingClock it sits directly below Network Password.
Live Form (RingClock only)	On/off toggle below NTP Polling Interval, On by default. When off, the RingClock will not serve the settings page in live mode; the only way back in is to hold SW2 for 5 s to return to AP setup mode.
NTP Polling Interval (minutes)	How often the clock re-syncs from the NTP server: 5, 15, 30, 60, 120 or 240 minutes. Does not affect boot or retry timing.
Firmware URL	Base URL for over-the-air updates. Default: https://mitchsf.github.io/Firmware/. RingClock accepts up to 128 characters.

Static IP Address (RingClock Only)

The Static IP Address section follows the network settings on the Setup page. With IP Address blank, the clock obtains its address automatically (DHCP), which is the default. A static address applies in live mode only; AP setup always uses 192.168.4.1.

Setting	Description
IP Address	Fixed address for the clock in live mode, up to 15 characters (for example 192.168.4.50). Leave blank for DHCP.
Gateway	Router address. Blank uses the IP Address with its last number replaced by 1 (shown as x.x.x.1).
DNS Server	Name server. Blank uses 8.8.8.8.

The subnet mask is 255.255.255.0. An entry that is not a valid address shows "Not a valid IP address (e.g. 192.168.4.50)"; an invalid entry is treated as blank when the clock starts.

DHCP fallback: if a boot with a static address cannot connect to WiFi and sync the time within the 2-minute boot budget, the clock restarts and the next boot uses DHCP once, so a wrong address still leaves the clock reachable. The boot after that tries the static address again. To find the clock after a fallback, use <device-name>.local or the router's DHCP table.

Time Zone (Both Clocks)

On the RingClock the time-zone settings are on the Setup page, under the UTC Offset, Daylight Saving Time and Standard Time headings. On the BitClock they are on the Time Settings page.

Setting	Options	Default
UTC Offset Hours	0–23	5
UTC Offset Minutes	RingClock 0–59; BitClock 0, 30, 45	0
UTC Offset Direction	Plus, Minus	Minus

Daylight Saving Time (Both Clocks)

Setting	Options	Default
DST Start Month	Jan–Dec	March
DST Start Week	1–5	2
DST Start Day	Sun–Sat	Sunday
DST Start Hour	0–23	2
DST Offset Hours (RingClock) / DST Offset (BitClock)	RingClock 0–13; BitClock 0–1	1
DST Offset Direction	Plus, Minus	Plus
ST Start Month	Jan–Dec	November
ST Start Week	1–5	1
ST Start Day	Sun–Sat	Sunday
ST Start Hour	0–23	2

Default settings are configured for US Eastern Time (GMT − 5 hr).

RingClock-Specific AP Settings

Setting	Description
Access Point / mDNS Name	Used for mDNS — the clock is reachable as <device-name>.local on networks supporting mDNS, when PSRAM is detected. Otherwise use its IP address.

BitClock-Specific AP Settings

Setting	Range	Default	Description
LED Length	33–500	33	Total number of LEDs on the strip. Must match the physical strip.

The setup LED Layout page contains LED Length, Power Supply Max (mA), and the four layout sliders. Enter the 5 V supply rating in milliamps; the clock limits LED power to that setting.

7. RingClock Display Modes

The RingClock's Display page exposes a Display Mode dropdown with eleven options (modes 0–9 plus Cycle). Each mode controls how the hour, minute, and second hands appear on the ring; the always-on layers (Pendulum, Pendant, 5-Minute Marks, Quarter Marks, Sparkle, Background, Drift) overlay on top of whichever mode is selected.

Modes

#	Display Mode	What it looks like
0	Dots	All three hands as single LEDs. Clean, minimal, easy to read at a glance.
1	Comet	Second hand sweeps with a fading comet tail; minute hand has a 3-LED trail; hour hand a single dot.
2	Blend	Default. Seconds fade into the current second; minutes hold until :55, then blend to the next minute by :00. Hour interpolates unless locked.
3	Orbit	Each hand sends a dot on one full orbit of the ring (1 second), painting a trail. All three hands then stop on the time for 4 seconds (the second hand keeps advancing) before orbiting again. The trails are not cleared; they fade as during the orbit.
4	Smooth Sweep	Hour and minute as dots; second hand moves continuously between adjacent LEDs for an analog sweep feel.
5	Radar	Faint ambient ring in the second-hand color with a bright 8-LED arc trailing behind the second hand — sonar-style.
6	Pulse	All three hands as dots, pulsing in unison once per second. Calm, ideal for dim rooms.

#	Display Mode	What it looks like
7	Fill From Hand	Each hand drawn as a 5-LED wedge centered on its position — bright in the middle, dimming outward.
8	Spark Chase	Each second a spark in the second-hand color sprints from 12 o'clock to the current second position with a cubic ease-out.
9	Gravity Drop	Each second a dot in the second-hand color drops from 12 o'clock with quadratic ease-in — slow start, accelerating finish, like gravity.
10	Cycle	Steps through modes 0–9 once a minute, a full cycle every 10 minutes. Starts at Dots after a reboot. Rules may supply a custom order with ao=.

Brightness

The Display Brightness slider (0–9) on the Display page sets overall brightness for the entire ring. The scale is gamma-corrected so each step feels perceptually even:

Level	Perceived Brightness
0	Off
1	Barely visible
2	Very dim
3	Dim
4	Low
5	Medium
6	Medium-bright
7	Bright
8	Very bright
9	Maximum

At low levels, animations may look slightly different and have a less smooth dimming gradient — a consequence of how addressable LEDs render small per-channel values.

Lock Hour

When On, the hour hand stays fixed at the exact hour position (12, 1, 2, 3 …) for the entire hour. When Off (default — the original BrightDot Clock behavior), the hour hand

advances one LED every 12 minutes, sliding gradually across the 5-LED span between hours. This mimics the behavior of a normal analog clock more closely.

Colors and Effects

The Display page has ten color pickers, a brightness slider beneath each hand color and each of the seven effect colors, and a Color Rotation toggle with a Rotation Period slider. Each color picker is a full RGB picker; setting any picker to all-zero black turns that element off.

Element	Default	Notes
Hour Hand	Red	Always rendered when brightness > 0
Minute Hand	Green	Always rendered when brightness > 0
Second Hand	Blue	Always rendered when brightness > 0
Pendulum	Purple #800080	Brightness 5 by default; slow simple-harmonic swing between LEDs 20 and 40; off when black
Pendant	Off (black)	Single dot bouncing the full ring; off when black
5-Minute Marks	Off (black)	12 marks; off when black
Quarter Marks	Off (black)	4 marks; layers on top of 5-minute marks
Sparkle Color	Off (black)	Random fading sparkles when non-black
Background Color	Off (black)	Dim wash across all 60 LEDs when non-black
Drift Color	Off (black)	8-LED CCW-drifting trail when non-black

Each hand has a brightness slider (1–9, default 9) beneath its color picker, relative to Display Brightness. At 9 the hands look exactly as before; a lower setting dims that hand alone but never turns it fully off — use a black color for that. Rules set them with hb=, mb= and sb=.

Each RingClock effect has a brightness slider (0–9) beneath its color picker. Pendulum defaults to 5; the other effect brightnesses default to 9, with their colors black (off). Brightness scales each layer relative to Display Brightness. A rule can set all seven together with eb=. Color Rotation moves every selected color through the hue spectrum over the Rotation Period (1–360 minutes, default 60).

Periodic Animations

Periodic Animation selects a five-second flourish that runs over the normal clock display. Animation Interval sets how often it runs: 15 seconds, 1, 5, 15 or 30 minutes, or Hourly. Factory defaults are Hand Spin every 1 minute. Display Mode continues to select the normal hand display between animations.

Periodic Animation / rule value	What it does
Off / pa=0	Disables periodic animations unless an active rule enables one.
Ripple / pa=1	Two pulses travel in opposite directions from the minute hand and meet across the dial.
Hour Gathering / pa=2	Twelve scattered lights gather at the current hour position.
Hand Spin / pa=3	The three hands circle the dial, then settle at their current time positions. Factory default.
Comet Pair / pa=4	Two comets circle in opposite directions with fading tails in the minute and second colors.
Color Tide / pa=5	A broad, dim wave in the minute color passes behind the hands.
Cycle / pa=6	Plays Ripple through Color Tide in order, one per interval, then repeats. Cycle starts at Ripple after a reboot or when selected.

Each periodic animation lasts five seconds and uses the current Display Brightness, including any active brightness rule. There is no separate animation brightness control. Pendulum remains a continuous effect in Colors and Effects.

Intervals follow clock boundaries: a 15-second interval runs at :00, :15, :30 and :45 (5 seconds of animation, then 10 seconds of clock); a 15-minute interval runs at :00, :15, :30 and :45 past the hour; Hourly runs at the top of the hour. Selecting an animation does not start it immediately. Wait for the next boundary. Booting or jumping the clock does not replay missed animations.

Changing display settings or applying rules cancels a running animation. Periodic animations are suppressed while brightness is zero or setup, IP display or firmware update owns the display. At completion, the normal clock resumes at the current time.

To run periodic animations only at selected times, set Periodic Animation to Off and save. Then add time-range rules with pa and ai as described in Section 10. A rule can choose the animation and interval independently of Display Mode.

8. BitClock Display

LED Layout

Four sliders on the setup LED Layout page tune the visual layout of the BitClock. The total width is computed automatically; any remaining LEDs become symmetric left/right margins. If the configured layout exceeds the strip length, it is clipped to LED Length.

Setting	Range	Effect
LED Data Bits	1–8	LEDs per bit cell (the visible width of a single 0 or 1)
LED Colon Bits	0–10	LEDs per colon lamp (two colons per clock face)
LED Bit Separator	0–8	Background LEDs between bits within the same digit
LED BCD Separator	0–10	Background LEDs between digits

The total LED count is set during AP configuration (Section 6) and constrained between 33 and 500.

Colors

The Time Display page has full RGB color pickers for FALSE, TRUE, COLON and BACKGROUND, each with its own brightness control from 0–9. Black turns that channel off. These brightness settings scale with the master brightness.

Channel	Description
TRUE Color	Lit bit cells (the 1s)
FALSE Color	Dim bit cells (the 0s)
COLON Color	The two colon groups between HH/MM/SS
BACKGROUND Color	Separators and margins

Setting any channel to black blanks it entirely. The most common pairings are TRUE = a saturated color, FALSE = black (so only the 1-bits are visible), but the firmware supports arbitrary combinations including matching TRUE and FALSE for a steady-glow look.

Time Display

Option	Description
12 Hour	Standard 12-hour clock
24 Hour	Standard 24-hour clock

Colon Display

Option	Description
Off	Colons always dark
On	Colons always lit in the COLON color
Blink	Colons alternate on/off once per second

Brightness

Ten levels (0–9), gamma-corrected.

Level	LED Value
0	Off
1	5
2	10
3	20
4	30
5	50
6	80
7	120
8	150
9	255

A single click of SW2 cycles brightness 1 → 2 → … → 9 → 1 (skipping 0) and persists the new value to flash immediately.

Date Display

On the setup Date Display page, choose Off, MM/DD/YY, DD/MM/YY or YY/MM/DD. Start Second (0–59) selects when the date appears each minute; Duration (0–15 seconds) controls the hold time. Transitions are Fade, Scroll, Split Scroll and Burst.

Date colors and brightness are independent of the time display: FALSE, TRUE, COLON and BACKGROUND each have an RGB picker and a 0–9 brightness slider. The live Date Display page enables or disables the date and tunes its timing, transition and colors. Set the date format in setup. The current firmware has no PIR motion-sensor controls.

9. Live Settings Web Interface

Once the clock is connected to WiFi, the live settings page is available from any device on the same network. All changes apply immediately as controls are adjusted — no reboot — and a Save action persists changes across power cycles.

Access

Browse to http://<clock-ip-address>/. On the RingClock, double-click SW1 to display the IP on the ring (see Section 12 for how to read it). On the BitClock, the IP is most easily found in the home router's DHCP table or through the .local hostname described next.

Using a Name Instead of an IP Address (mDNS)

On home networks that support mDNS (also called Bonjour or zero-configuration networking), the clock can be reached by name. The hostname is derived from the Access Point / mDNS Name — lowercased, with spaces and any non-alphanumeric characters converted to hyphens. Defaults are http://ringclock.local/ and http://bitclock.local/. The same name is sent as the DHCP client hostname, so the clock is identifiable in the router's connected-device list too. Both clocks enable mDNS only when PSRAM is detected; without PSRAM, use the IP address.

mDNS does not work on every network. Some routers — particularly older models or routers with multicast filtering enabled — block the discovery messages mDNS depends on, in which case the .local address will not resolve. macOS, iOS, and Windows 10/11 generally work out of the box; some Android versions and Linux distributions need extra setup. If the .local address does not load, fall back to the IP address.

Adding the Settings Page to a Phone Home Screen

The live settings page on either clock can be installed on a phone or tablet like an app, giving it its own home-screen icon that opens straight to the settings with no browser address bar. Open the live page, go to the System page, and tap Add to Home Screen. On Android the browser normally offers to install it directly; on an iPhone or iPad, tap the Share button in Safari and choose Add to Home Screen, which the button will display as a reminder.

The icon is labelled with the Access Point / mDNS Name, so several clocks on the same phone each get their own clearly-named icon. Because the icon remembers the exact address it was installed from, install it from the .local name where mDNS works — an icon saved against a DHCP address stops working if the router later hands the clock a different one.

Live Form Password

If a Live Form Password was set in AP configuration, the page prompts for it before any settings appear. Leave the password field blank in AP configuration to disable password protection.

Saving Changes

Adjustments take effect on the display the moment a control is released — useful for trying out colors, display modes, or layout without commitment. To make changes permanent, tap Save at the bottom of the page; the clock writes the configuration to flash so it survives a power cycle. Reloading the page without saving leaves the running settings in place but does not persist them. Rule edits, deletions and ordering changes save immediately; they do not require Save.

Pages and Controls

RingClock live mode has Display, Rule Editor and System pages. The System page offers Firmware URL, Update Firmware, Automatic Updates, Switch to AP Mode and Add to Home Screen. Setup mode instead has Setup and Display, plus Start; the Rule Editor is available only after joining WiFi. BitClock live mode has Time Display, Date Display, Rule Editor and System pages. Its setup pages are Setup, Time Settings, LED Layout, Time Display and Date Display.

10. Scheduled Rules

The Rule Editor replaces the old three-period Overrides page. Rules combine a time window with selected display settings. On both clocks, the top matching rule takes priority when rules set the same field, and rules setting different fields combine.

Editing RingClock Rules

Open Rule Editor in live mode. Enter a rule and press Apply; tap a schedule row to edit it. Delete removes the selected row, and the up/down arrows change priority. Edits, additions, deletions and moves save immediately, so this page has no Save button. Each change is confirmed with a message: Rule added, Rule updated or Rule deleted. The schedule shows every rule in one table. The maximum number of rules depends on the model and firmware version. When the list is full, the Rule Editor refuses a new rule. If a saved schedule is longer than the limit (for example after a firmware update), only the rules at the top of the list are kept.

Rules Enabled is a temporary switch that returns to On after reboot. Manually changing a display setting while a rule is active disables Rules. Changing the Firmware URL or Automatic Updates on the System page leaves Rules on. Re-enable the switch to resume the schedule.

Time Windows

Times have minute precision. The end is exclusive: t=22:00-7:00 runs from 10 PM through 6:59 AM. Cross-midnight windows are supported; equal endpoints make a window inactive. Add wd or we for weekdays or weekends. Commas or whitespace separate fields. The Rule field tooltip lists the accepted syntax.

RingClock Fields

Keys	Meaning
br, dm	Display Brightness 0–9; Display Mode 0–10, 10 = Cycle (dm or displaymode)
pa	Periodic Animation: 0 Off, 1 Ripple, 2 Hour Gathering, 3 Hand Spin, 4 Comet Pair, 5 Color Tide, 6 Cycle (pa or periodicanimation)
ai	Animation Interval in seconds: 15, 60, 300, 900, 1800 or 3600 (ai or animationinterval)
iv, lh	Color inversion; lock hour (0/1)
hc, mc, sc	Hour, minute and second colors
hb, mb, sb	Hour, minute and second brightness, 1–9
pn, pd, fc, qc	Pendulum, pendant, five-minute and quarter-mark colors
sp, bg, dr	Sparkle, background and drift colors

Keys	Meaning
eb	Set all seven effect brightnesses together, 0–9
cr, rp	Color rotation (0/1); rotation period 1–360 minutes
ao	Custom Cycle order of display modes 0–9; implies Cycle (dm=10) (ao, animationorder or anorder)

Example: t=22:00-7:00 br=2 dims the clock overnight. Example: t=18-23 pn=#800080 eb=5 enables a purple pendulum and sets all seven effect brightnesses to 5 in that window. Unspecified settings continue to use the Display defaults.

Periodic animation examples: t=8-12 pa=3 ai=900 runs Hand Spin every 15 minutes in the morning; t=12-18 pa=1 ai=300 selects Ripple every five minutes in the afternoon; t=18-22 pa=4 ai=1800 selects Comet Pair every half hour in the evening. Set the Display default to Off if no periodic animations are wanted outside those windows. To test Ripple every 15 seconds, use t=all pa=1 ai=15. The interval is a number of seconds, not a dropdown index.

Use dm for Display Mode, pa for Periodic Animation, ai for Animation Interval and ao for a custom Cycle order. The long names displaymode, periodicanimation, animationinterval and animationorder are also accepted. A rule that sets ao with a Display Mode other than Cycle is rejected. Periodic Animation settings and rules saved by older firmware, before the current Periodic Animation numbering, are not converted; re-select the animation and re-enter such rules.

BitClock Fields

BitClock also uses a Rule Editor. Its fields differ: br for master brightness, cd for colon display, tf for time format, fc/fb for false-bit color/brightness, tc/tb for true-bit color/brightness, cc/cb for colon color/brightness, bc/bb for background color/brightness, and ds/du/tr for date start second, duration and transition. Use its on-device tooltip for the valid ranges. Its schedule also shows every rule in one table, and the same rule limit applies: the maximum number of rules depends on the model and firmware version.

11. NTP Time Sync

Both clocks obtain the current time from an internet NTP (Network Time Protocol) server. After WiFi associates on boot, the clock contacts the configured NTP server and uses the result to set its internal clock. Daylight Saving Time is then applied automatically from the configured DST rules.

Choosing an NTP Server

RingClock defaults to us.pool.ntp.org; BitClock defaults to time.cloudflare.com. Both use time.google.com as the alternate. RingClock offers polling intervals of 5, 15, 30, 60, 120 and 240 minutes, defaulting to 240. Choose an interval appropriate to the time-server provider.

Local NTP Servers

A single clock polls infrequently, but a household with many clocks all pointed at the same public pool adds load to a volunteer-run service. In that case, point each clock at a local time server instead. Angry Electrons offers such a device for sale that uses GPS to sync an internal time server for use on a home network.

A router or local NTP appliance can serve time if it explicitly supports NTP. Use its actual configured address; do not assume that the router itself is a time server.

Alternate NTP Server (Both Clocks)

Both clocks can be configured with a primary and an alternate NTP server. If the primary fails to respond for 30 minutes, the firmware silently rotates to the alternate so the clock keeps syncing through transient outages. Leave the alternate field blank to disable rotation.

12. Showing the IP Address (RingClock)

Double-click SW1 on the RingClock to show the clock's IP address on the ring. Each digit is rendered as that number of lit LEDs. Within each octet, digits are colored red (first), green (second), and blue (third). Octets are separated by a single white LED.

A single click of SW1 dismisses the display and returns to the configured time mode. The display stays until dismissed, so there is time to read it from across the room.

If the IP is too long to fit on the ring (more than 55 LEDs needed), the display splits into two pages: the first two octets for 10 seconds, then the last two for 10 seconds, cycling until SW1 is clicked.

The same dot-string format shows the firmware version for five seconds on ordinary RingClock boots. Silent automatic-update boots skip it. An ordinary boot also runs a red/green/blue self-test at 25% peak; the version and connection sweeps use 40% brightness.

13. Over the Air Update

Both clocks can update their software over WiFi — no USB cable, no programming tools required. To trigger an update, hold SW1 for 10 seconds while the time is displayed. The clock contacts the firmware server, compares versions, downloads the new image if needed, flashes it, and reboots.

Default URL: https://mitchsf.github.io/Firmware/. The clock fetches a small VERSION file from this URL to determine the latest available release; if the server reports a newer version, the corresponding .bin file is downloaded and flashed.

RingClock Display Feedback

Manual RingClock OTA status uses 50% brightness and the install-orientation 12 o'clock position. After Update Firmware is pressed on the System page, the browser shows "Checking for a firmware update…", then "updating to v<version>…" while a validated target version is announced on the ring for about four seconds before downloading, or "firmware already current (v<version>)". Progress is the actual 000–100 percentage in three 4-LED BCD groups, separated by black LEDs: set bits use the hour-hand color and clear bits use the minute-hand color.

Display	Meaning
Empty ring	Connecting to WiFi and the firmware server
Target version (about 4 s)	New version announced before manual download
Version dot-string (5 s)	Software is already current. Returns to normal time display after 5 seconds.
Three BCD groups, 000–100	Actual download percentage using hour/minute colors for set/clear bits
All 60 LEDs solid green (5 s)	Update successful. Reboots automatically.

RingClock Error Codes

On failure, a small bar of red LEDs appears at the 12 o'clock position for 5 seconds, then the clock reboots:

LEDs lit red	Meaning
1	No WiFi connection or firmware URL not configured
2	Could not fetch the VERSION file from the server
3	The VERSION file is invalid
4	Software download or flash write failed

BitClock Display Feedback

OTA status renders as two-digit BCD codes — each digit drawn as a 4-LED group (ON = 20% green, OFF = 20% red), with a 1-LED black gap between groups. The pattern mirrors the Zev-7 nixie display codes:

Display	Meaning
88	Connecting to WiFi and the firmware server
Version (e.g. 1 0 7) for 3 s	Software is already current. Returns to normal time display after 3 seconds.
Left digit cycles 0–9, right blank	Downloading and flashing new software (counter ticks every 200 ms)
10	Update successful. Reboots automatically after 3 seconds.

BitClock Error Codes

On failure, the strip displays a two-digit error code starting with 9 for 5 seconds, then the clock reboots:

Code	Meaning
91	No WiFi connection or firmware URL not configured
92	Could not fetch the VERSION file from the server
93	VERSION file is malformed (missing or invalid version string)
94	Software download or flash write failed

Version Behavior (Both Clocks)

Firmware updates preserve settings, including major-version updates. Only an explicit factory reset or first setup applies the factory defaults. Updating an existing RingClock does not overwrite its saved pendulum color, brightness or Display Mode (a removed display mode becomes Blend). Periodic Animation choices saved by older firmware, before the current numbering, are not converted; re-select the animation after updating.

RingClock Automatic Updates is On by default. It checks weekly at a fixed per-device time between 2:30 and 3:30 AM and installs only the release approved by the matching Auto=<version> manifest entry. These updates are silent; failures resume normal operation. Automatic-update boots skip the self-test and version display.

14. Troubleshooting

Symptom	Possible Cause	Solution
Clock starts in AP mode unexpectedly	Settings cleared or factory reset triggered	Reconfigure WiFi at 192.168.4.1 and press Start. If this happens after every boot, check that both buttons are not stuck physically pressed.
Clock starts in AP mode after a WiFi password change or network rename	Three boots in a row failed to connect to WiFi and sync the time	Connect to the setup AP, correct the SSID and password at 192.168.4.1, and press Start. If the network was only temporarily down (router reboot or internet outage), power-cycle the clock instead; it retries the saved network.
Cannot connect to RingClock / BitClock WiFi	Clock is not in AP mode	RingClock: release SW2 after 5–10 seconds. BitClock: hold 5 seconds. Both buttons at power-on reset settings.
Boot completes but no time appears	WiFi associated but NTP failed	Verify the configured NTP server is reachable. Try time.cloudflare.com or time.google.com. If WiFi and NTP have not both completed after two minutes, the clock auto-reboots.
RingClock numbers read backwards or upside-down	Install orientation mode does not match physical mounting	Power-cycle with SW2 held to advance to the next orientation mode. Cycle through the five install modes to find the correct combination (Section 5).
BitClock layout is clipped or off-center	LED Length set incorrectly, or layout sliders too large for the strip	Confirm LED Length matches the physical strip in AP setup. Reduce the four layout sliders if the right edge is clipped.
Live page does not load	Live Form Password set, or wrong IP address	Confirm the password if one was set. Use mDNS (<ap-name>.local) or the home router's DHCP table to find the correct IP.

Symptom	Possible Cause	Solution
mDNS .local address does not resolve	Clock has no PSRAM, or network/device blocks mDNS	Fall back to the IP address. macOS, iOS, and Windows 10/11 generally work; older Android and some Linux distros need extra setup.
Scheduled rules never take effect	Time-zone or DST misconfigured, so the local clock never enters the period's window	Check Rules Enabled, local time, day filters, rule times and priority.
Over-the-air update shows red LED(s) (RingClock) or 9X code (BitClock)	No WiFi, server unreachable, or download failed	Check WiFi association. Confirm the firmware URL is correct and the server is reachable from another device on the same network.
5 GHz network is not in the SSID dropdown	ESP32 hardware only supports 2.4 GHz	Connect the clock to a 2.4 GHz network. Most home routers broadcast both — select the 2.4 GHz one.

Notices

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Operate the clock at the supplied 5 V DC voltage only; higher voltages will damage the LEDs. The RingClock and BitClock do not contain user-serviceable parts.

RingClock & BitClock — Designed by Mitchell Feig